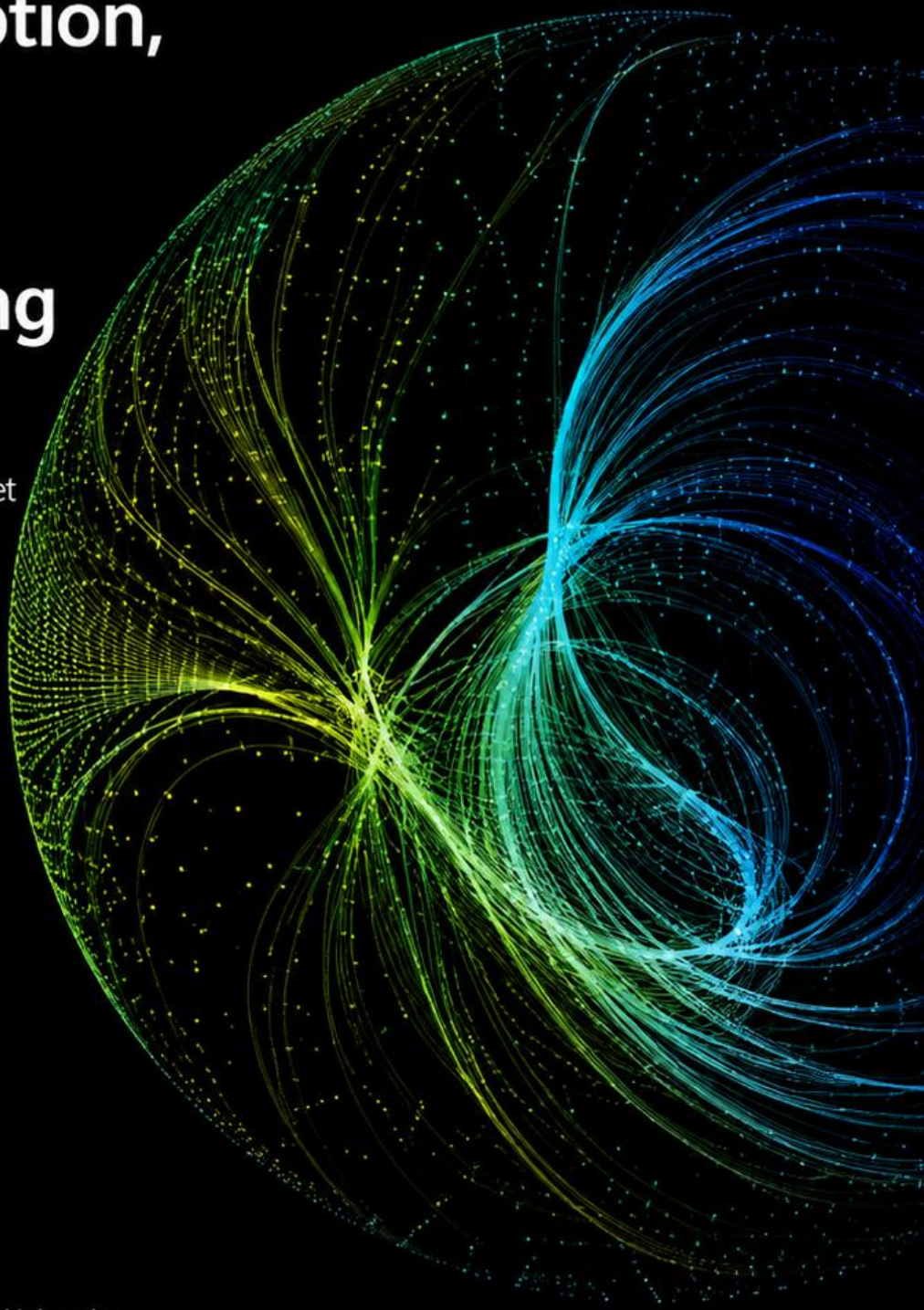
M2 THESIS | TECHNICAL REPORT

Data Description, Exploratory Analysis & Preprocessing

IEEE-CIS Fraud Detection Dataset



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Comparative Feature Selection and Predictive Models for E-Commerce Fraud Detection

01 Executive Summary

This report documents the descriptive analysis and preprocessing pipeline applied to the IEEE-CIS Fraud Detection dataset ahead of the comparative feature-selection and classification study. Due to its scale, high dimensionality, severe class imbalance, and temporal nature, this dataset serves as a realistic benchmark for evaluating fraud detection methods. This document summarizes the dataset's core characteristics, evaluates temporal properties under a chronological split, and details a leakage-aware preprocessing protocol used to generate model-ready datasets for the subsequent machine learning experiments.

590K	434	3.5%	27.6:1
Transactions	Raw features	Fraud rate	Class ratio

Research objective: The primary objective is to identify preprocessing and feature selection strategies that maximize predictive performance while minimizing data dimensionality.

Dataset scale: The dataset merges transaction and identity tables via TransactionID, yielding 590,540 labeled transactions across 434 variables. Fraudulent transactions account for approximately 3.5% of the observations, reflecting a severe class imbalance.

Report structure: I will describe the dataset in section 2, Section 3 presents the exploratory data analysis (EDA), focusing on missingness and distributional shapes. Section 4 details the step-by-step preprocessing pipeline and data-splitting strategy.

02 Dataset Description

The **IEEE-CIS Fraud Detection** dataset was released during a Kaggle competition organized by the IEEE Computational Intelligence Society and provided by **Vesta Corporation**, an e-commerce payment fraud prevention specialist. Representing **card-not-present (CNP)** transactions, the data consists of two relational tables: a transaction table containing payment features and an identity table containing device and network attributes, linked via the **TransactionID** variable. Because identity logs are only available for a subset of events, the tables were combined using a left join. This approach retains all transaction records while naturally treating unmatched identity attributes as missing data.

Component	Observed size / role	Content
Transaction data	590,540 rows × 394 columns	Transaction amount and timing, product and card attributes, addresses, email domains, count/time/match variables, Vesta-engineered variables, and the target isFraud.
Identity data	144,233 rows × 41 columns	Device metadata, software profiles, network routing configurations, and identity-verification variables collected exclusively for a transaction subset.

Feature families

Variable group	Meaning
TransactionID	Unique transaction identifier (dropped from the predictor set).
TransactionDT	Continuous time delta recording elapsed seconds from an unrevealed baseline date; utilized for chronological validation partitioning and retained as a candidate predictor.
TransactionAmt	Transaction amount in US dollars.
ProductCD	Categorical product code capturing the high-level transactional domain (e.g., online, cloud, or physical goods).
card1–card6	Anonymised payment-card attributes (type, category, issuing bank, country, etc.).
addr1–addr2; dist1–dist2	Anonymised address and distance-related information.
P_emaildomain; R_emaildomain	Purchaser and recipient email domains.
C1–C14	Anonymised count/frequency variables (e.g. addresses associated with a card); exact meaning masked.
D1–D15	Anonymised time-delta variables (e.g. days since previous transaction).
M1–M9	Boolean match indicators (e.g. name-on-card vs address).
V1–V339	Vesta-engineered anonymised variables (rankings, counts, entity relations).

03 Exploratory Data Analysis

Exploratory data analysis (**EDA**) was conducted to examine the principal characteristics of the **IEEE-CIS Fraud Detection dataset** before model development. The analysis focuses on class imbalance, missing-value patterns, feature distributions, temporal variation, and potentially informative categorical groups. Descriptive analyses are performed on the complete labelled dataset to provide an overall understanding of the data; however, all preprocessing operations and predictive models are subsequently developed using only the chronological training partition to prevent information leakage.

3.1 Class imbalance

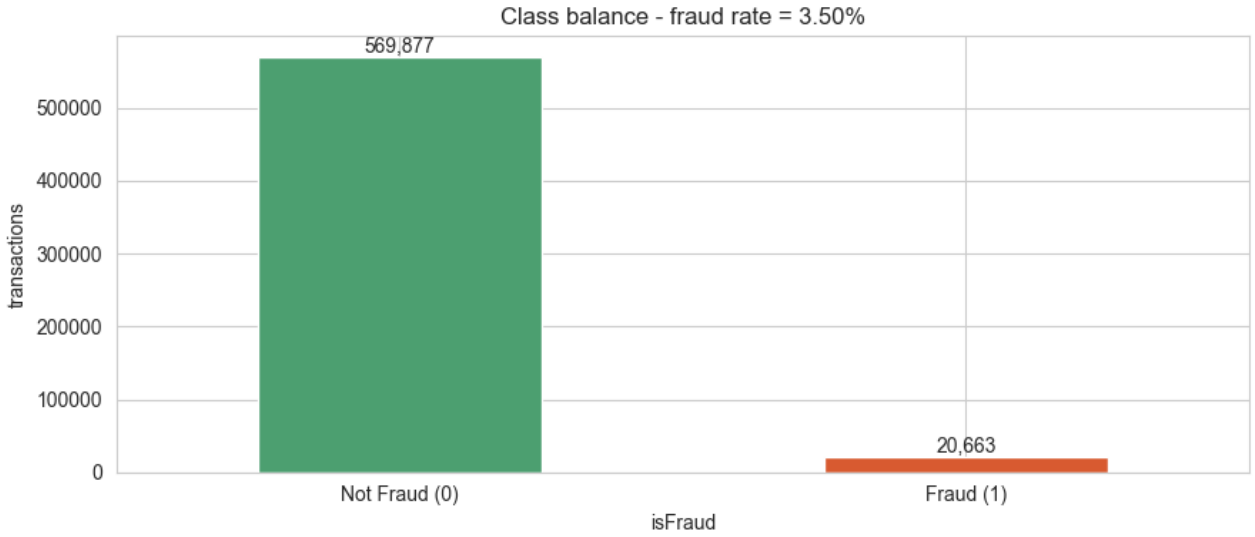


Figure 1. *Target distribution. Fraud accounts for 20,663 of 590,540 transactions (3.5%).*

The target variable, isFraud, is severely imbalanced: it consists of 569,877 legitimate transactions (96.5%) against 20,663 fraudulent ones (3.5%). This severe class imbalance represents one of the principal challenges of e-commerce fraud detection. Because fraudulent transactions constitute only a small fraction of all observations, standard learning algorithms risk optimizing for the majority class, making traditional metric evaluations like overall accuracy highly deceptive.

> **Implication** Cost-sensitive class weighting in every model, and evaluation on Precision-Recall Curve (PR-AUC), Receiver Operating Characteristic (ROC-AUC), and MCC rather than accuracy, which is uninformative at this prevalence.

3.2 Missingness

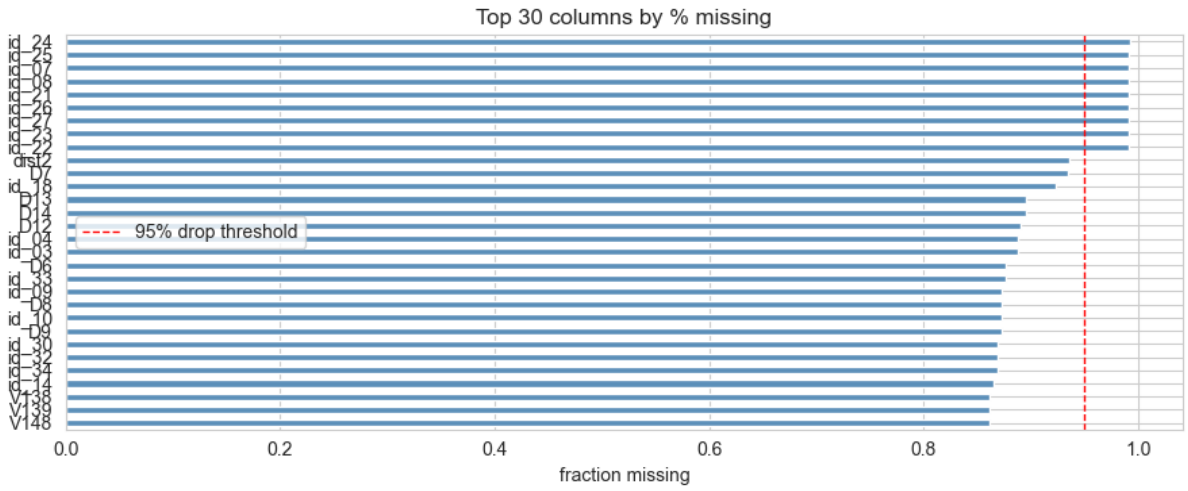


Figure 2. *The thirty columns with the highest missing-value proportions; nine exceed the 95% threshold (dashed line).*

Most of the variables with the highest missing rates belong to the identity table, with several exceeding 99% missing values (id_24, id_25, id_07, id_08, id_21, id_26, id_27, id_23, id_22). This is expected because identity information is available for only about **144,000** of the **590,540** transactions, leaving the remaining records completely unmapped post-merge. Crucially, in e-commerce fraud modeling, missingness is often non-random (Missing Not at Random, or MNAR) and can serve as a strong predictive signal of fraudulent behavior.

> **Implication** Rather than arbitrarily dropping these high-missingness variables, they are preserved for downstream modeling.

3.3 Transaction amount

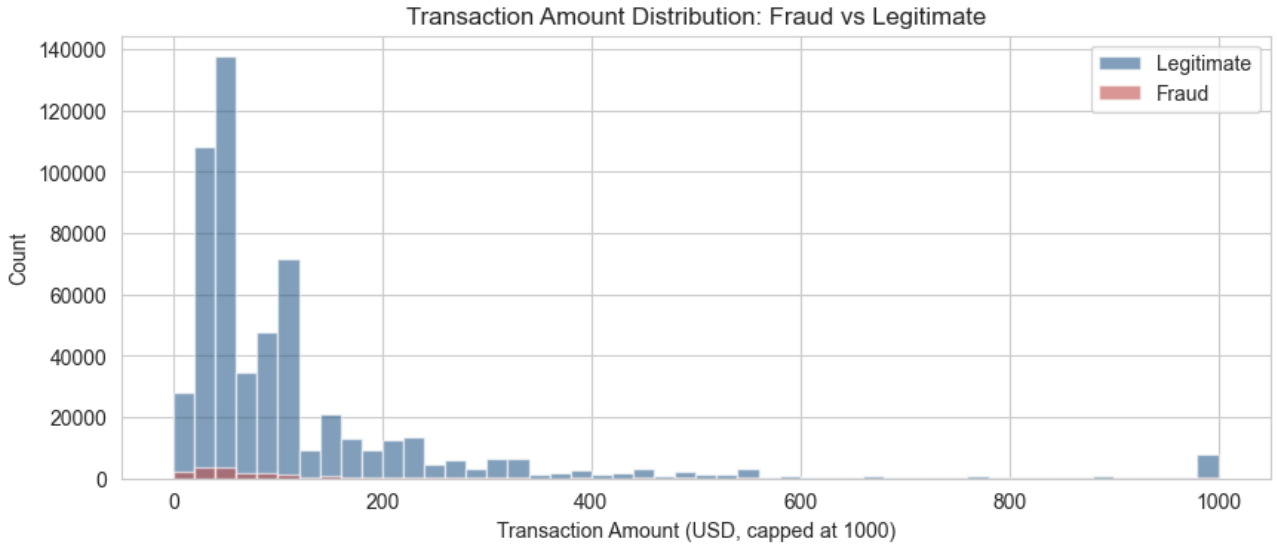


Figure 3. *Distribution of transaction amounts for legitimate and fraudulent transactions. the maximum observed value is ≈ USD 31,937.*

Transaction amounts are strongly right-skewed for both legitimate and fraudulent transactions, with most observations concentrated below approximately USD 200. Although fraudulent transactions occur across a wide financial range, their distribution largely overlaps with that of legitimate transactions, indicating that transaction amount alone cannot reliably distinguish fraudulent behavior.

Implication Use the median, not the mean, for numeric imputation. The observed skewness is typical of financial transaction data and is expected to be handled effectively by the machine learning models evaluated in this study.

3.4 Temporal structure

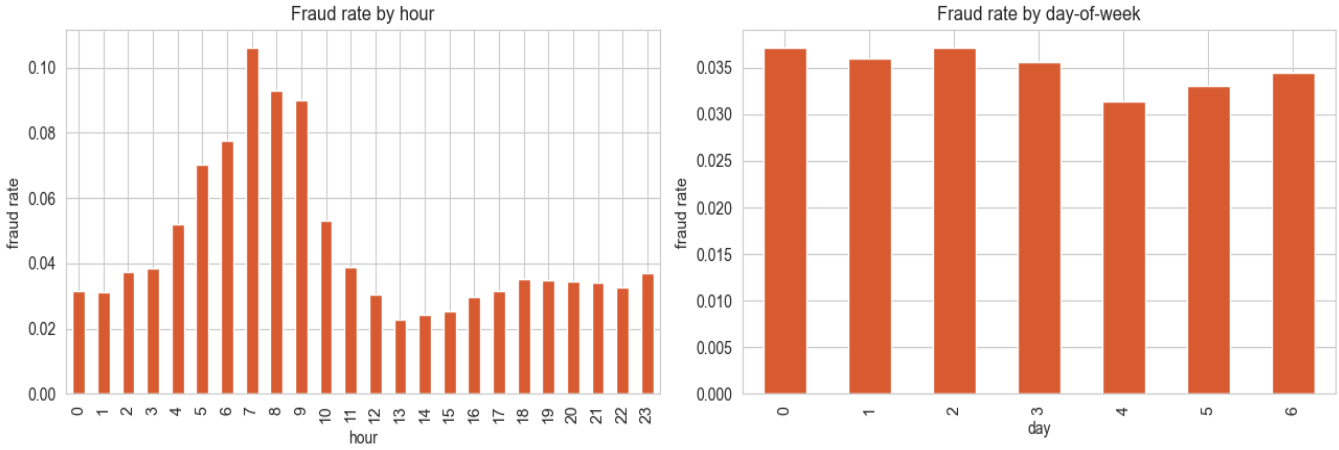


Figure 4. *Fraud rate by derived hour (left) and day-of-week (right).*

The dataset's temporal structure is engineered via the TransactionDT variable, which records the time delta in seconds from an unrevealed baseline date. Engineering cyclical time components reveals that fraud rates vary distinctively across a 24-hour cycle, peaking significantly during the early morning hours, specifically around 07:00, the fraud rate decreases and remains fairly stable. Conversely, minimal variance in fraud rates is observed across the days of the week, indicating that diurnal fluctuations contain higher predictive utility than weekly cycles.

Implication A strict chronological split is used to divide the training and testing partitions. This approach prevents data leakage and ensures that the model evaluation mimics a realistic deployment scenario, where estimators must predict future fraud using only historical data.

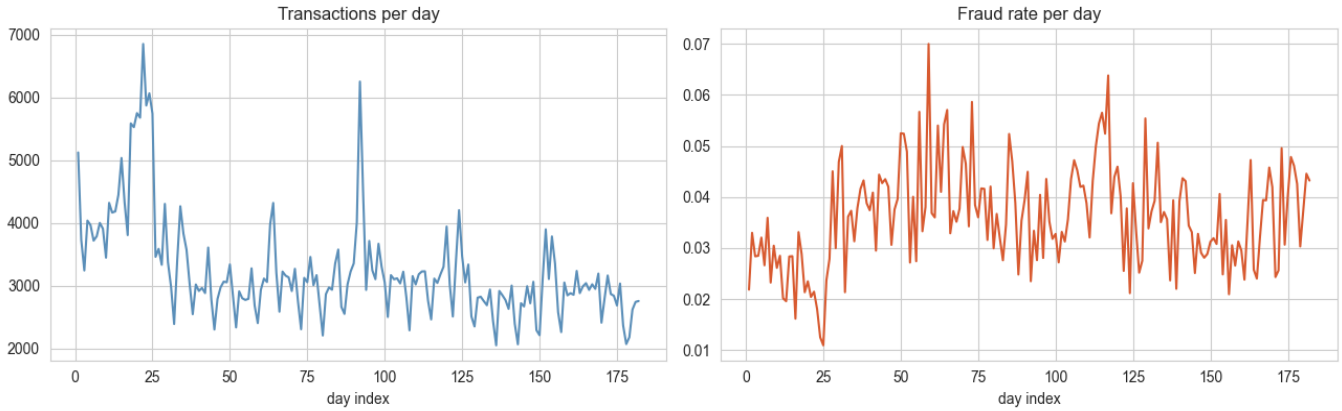


Figure 5. *Daily transaction volume (left) and daily fraud rate (right) across the observation window.*

The daily transaction volume exhibits considerable volatility over the observation period, characterized by distinct periodic peaks and drops rather than a stationary baseline. Concurrently, the daily fraud rate experiences substantial fluctuations over time, typically varying between 2% and 6%, with a few days showing higher values.

> **Implication** preserving the chronological order of the data provides a more realistic evaluation than a random split, as it better reflects the practical deployment of fraud detection models on future unseen transactions.

3.5 Categorical fraud concentration

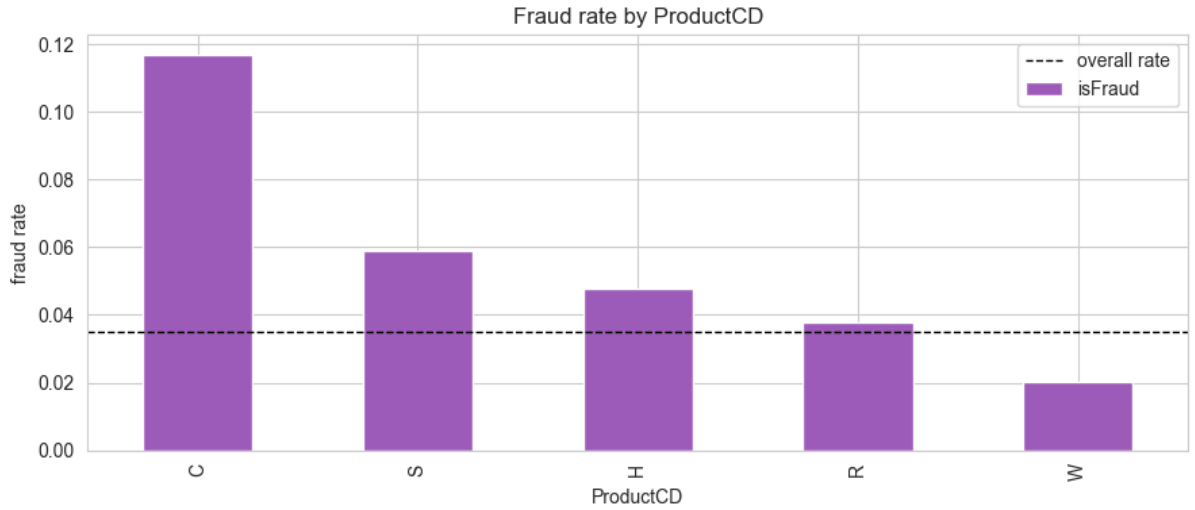


Figure 6. *Fraud rate by product code (ProductCD).*

The fraud rate varies significantly across the five product codes, revealing strong categorical associations with the target variable. Product Category C (typically representing online/cloud products) exhibits the highest risk profile, with a fraud rate of approximately 11.7%, more than triple the dataset's global average of 3.5%. Categories S and H also display elevated, above-average risk profiles, whereas Category W presents the lowest relative fraud incidence.

> **Implication** These stark distributional disparities confirm that product type is a critical determinant of transaction risk. Therefore, the ProductCD feature possesses high mutual information with the target variable.

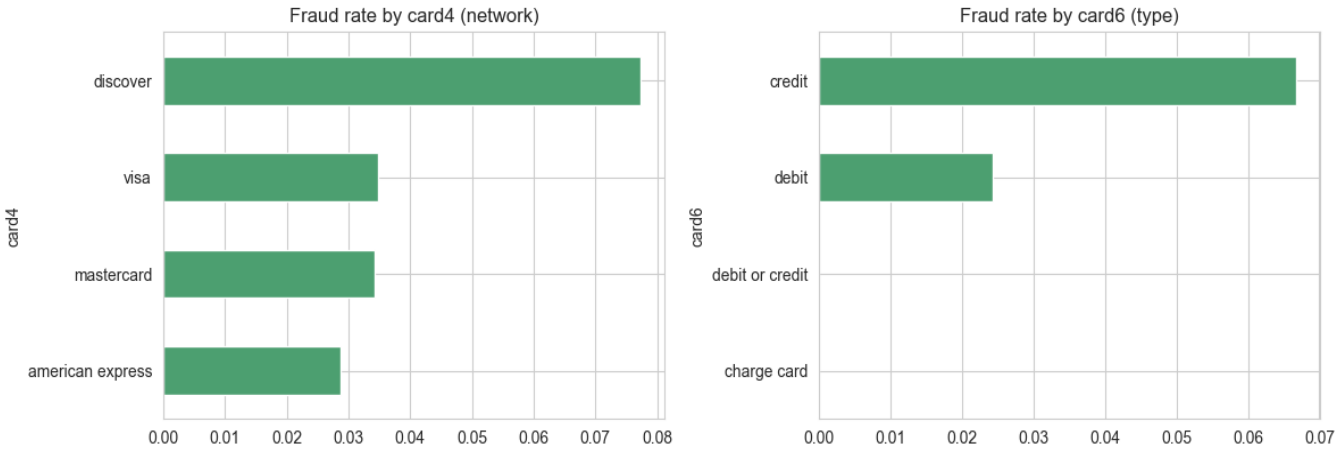


Figure 7. *Fraud rate by card network (card4) and card class (card6).*

Fraud rates vary across both card networks and card types. Among the primary payment networks, **Discover** registers the highest relative fraud incidence, whereas **American Express** demonstrates the lowest baseline risk. **Visa** and **Mastercard** display tightly coupled fraud rates that mirror the dataset's global average. A more pronounced structural variance is observable when analyzing the transaction instrument class (card6), where **credit** instruments present a significantly higher fraud rate than **debit**

instruments. This disparity underscores the higher inherent risk associated with credit transactions in card-not-present environments. Ultimately, both features provide orthogonal predictive signals regarding cardholder risk profiles.

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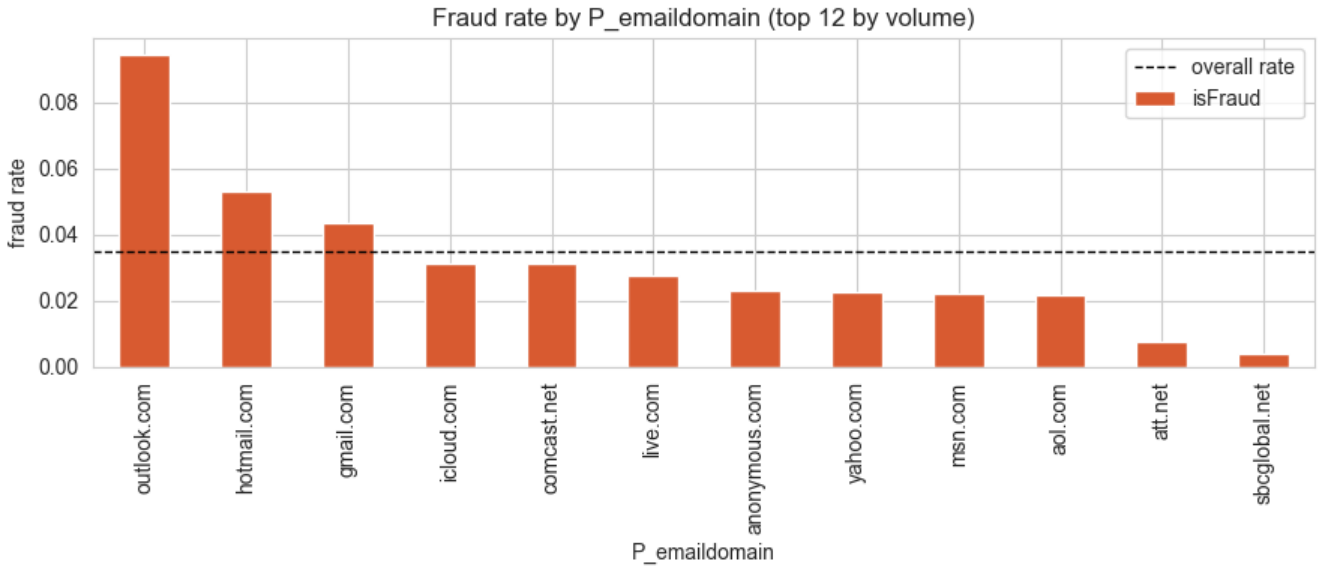


Figure 8. *Fraud rate by purchaser email domain (P_emaildomain), for the most common domains.*

The fraud incidence varies across the most common purchaser email domains; specifically, transactions associated with outlook.com exhibit an elevated risk profile that significantly exceeds the dataset's global average. Conversely, several alternative mainstream domains remain near or below the baseline fraud rate. This variance confirms that email domain registration serves as a strong proxy for transaction legitimacy.

> **Implication** Suggesting that P_emaildomain may contain useful information for distinguishing fraudulent and legitimate transactions.

3.6 Redundancy among engineered variables

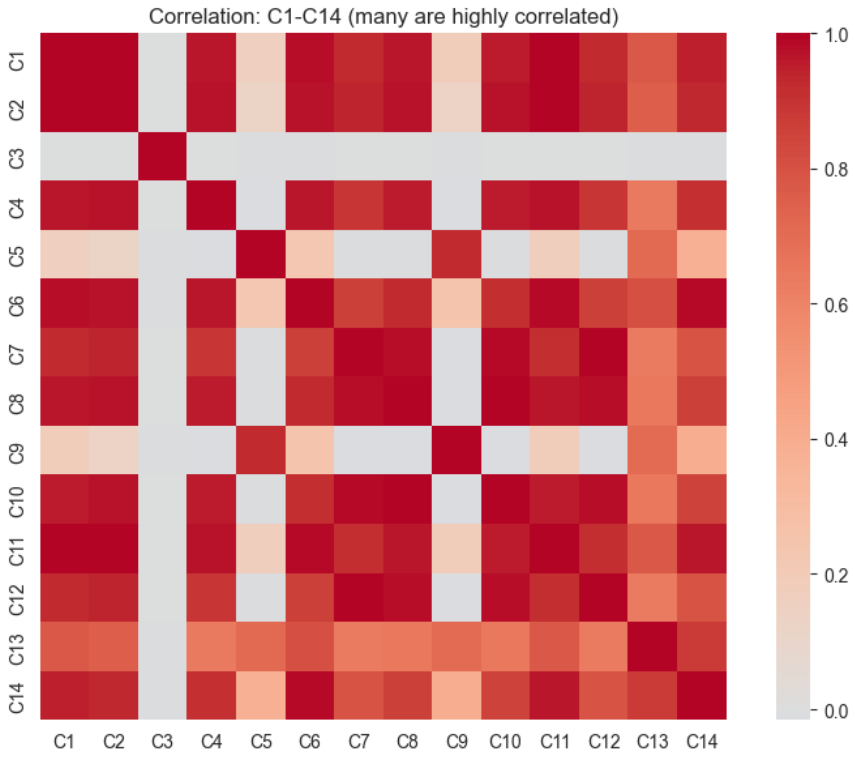


Figure 9a. *Correlation among the count variables C1–C14.*

Most count variables are strongly positively correlated, suggesting that they contain overlapping information. C3 is a clear exception, showing little correlation with the other variables and therefore capturing different information. This redundancy may affect linear models more than tree-based models such as XGBoost and LightGBM, which are generally less sensitive to correlated inputs.

> **Implication** This structural redundancy raises the question the comparative study is designed to answer: whether removing correlated features improves predictive performance, or whether the models already handle redunancy internally.

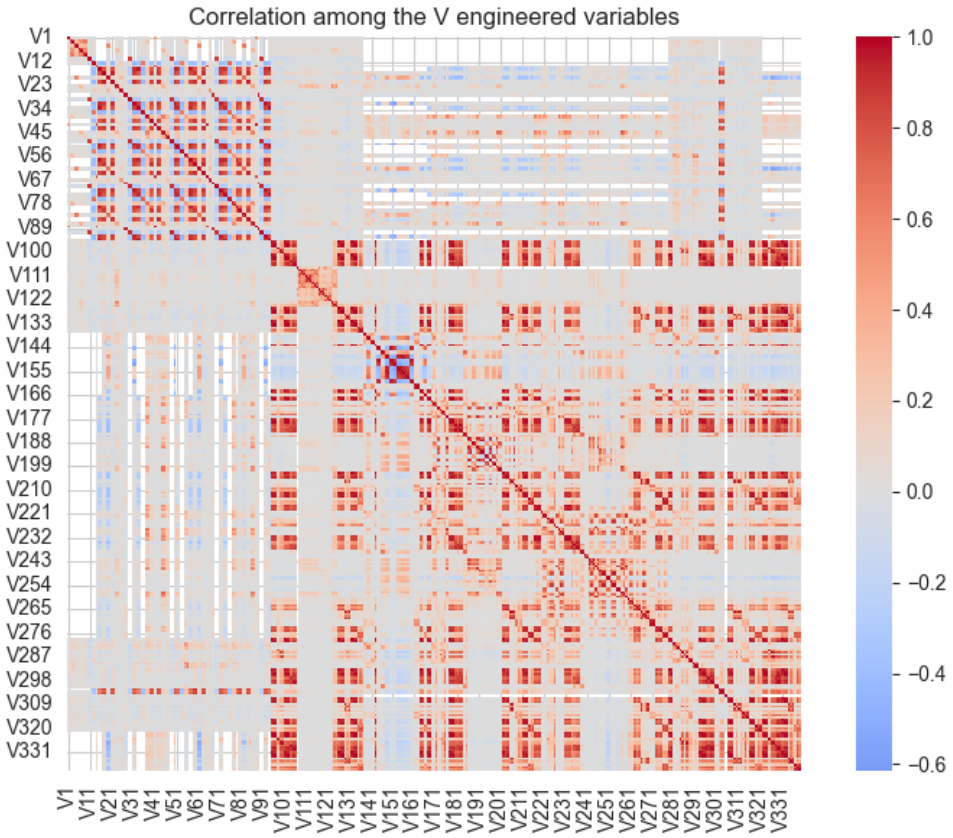


Figure 9b. *Correlation among the V1–V339 engineered variables.*

The V variables show a clear cluster structure, with some groups of features being highly correlated while relationships between different groups are generally weak. However, the overall level of redundancy is moderate: the mean absolute correlation is 0.153, only 2.6% of feature pairs have correlations above 0.90, and 104 of the 339 principal components are required to explain 95% of the variance. This suggests that, although some V variables contain overlapping information, the block still contains substantial variation across its features.

> **Implication** This correlation structure may affect the models differently. Tree-based models can select among correlated features during splitting, while linear models can be more sensitive to multicollinearity. L1 regularizations may also retain one feature while excluding other highly correlated features. This provides an additional reason to evaluate feature selection across different classifier families.

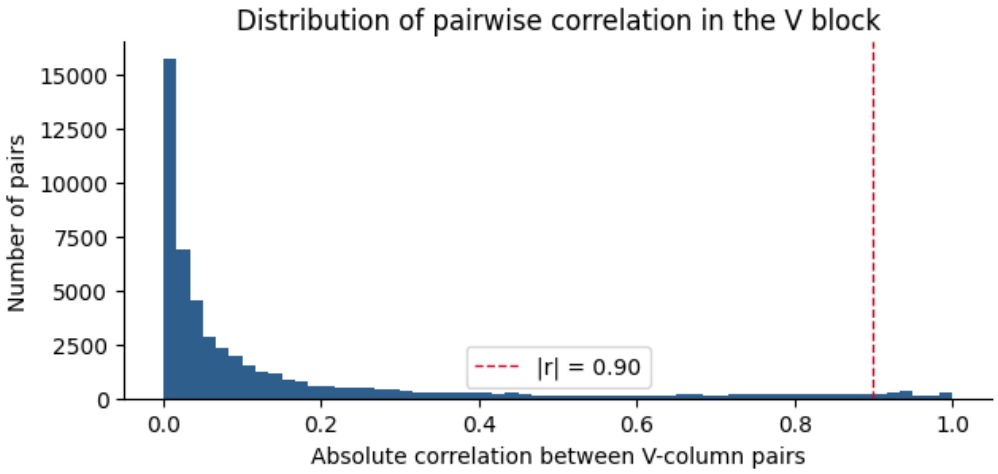


Fig.10 *Distribution of absolute pairwise correlation among the V engineered variables.*

The distribution is heavily concentrated near zero. The mean absolute correlation across the 51,731 pairs is 0.153, and only 2.6% of pairs exceed 0.90, with 139 pairs above 0.99. The small mass in the right tail corresponds to the dense blocks visible in Figure 9b, where small groups of variables are near-duplicates of one another. The V block is therefore not broadly redundant. It contains isolated pockets of near-duplication within a feature space whose variables mostly carry distinct information, meaning its effective dimensionality is close to its column count rather than far below it.

> **Implication** The V variables show partial redundancy rather than complete duplication. Therefore, reducing this feature block may remove useful information, and its effect on predictive performance needs to be tested empirically.

3.7 Training vs test period after the chronological split

The following figures compare the **training dataset** (comprising the first 80% of the transactions) with the **test dataset** (the remaining 20%). Because the data were divided chronologically, the comparison focuses on how the characteristics of the transactions change over time rather than on differences caused by random sampling. The objective is to identify any changes in the distributions of the main variables between the two periods. Given the large size of the dataset, the comparison is based on visual inspection and the magnitude of the observed differences rather than statistical significance tests, which can detect very small differences that are not practically important.

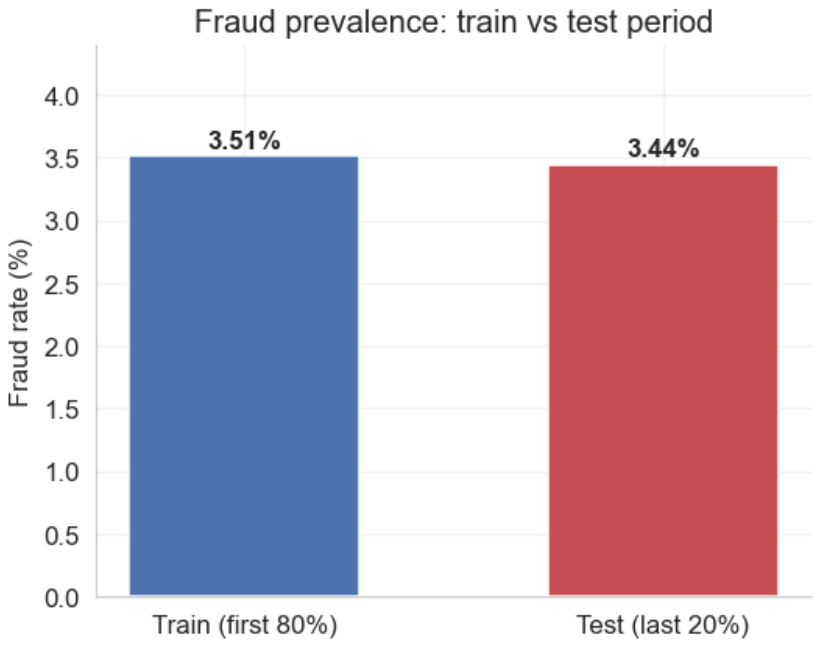


Figure 11. *Fraud prevalence: training (earlier 80%) vs test (later 20%).*

The baseline fraud prevalence is highly consistent across the chronological split, registering at **3.51% in the training** partition and **3.44% in the test** partition, representing a minor relative variance of approximately 2%. In highly imbalanced contexts, the empirical base rate defines the random-guess baseline under the Precision-Recall curve (AUC-PR). This stability ensures that the downstream evaluation AUC-PR scores are directly comparable to training expectations and are not artificially altered by a shift in the target class distribution.

> **Implication** This stability satisfies a critical precondition for reliably interpreting downstream classification performance across the two temporal windows.

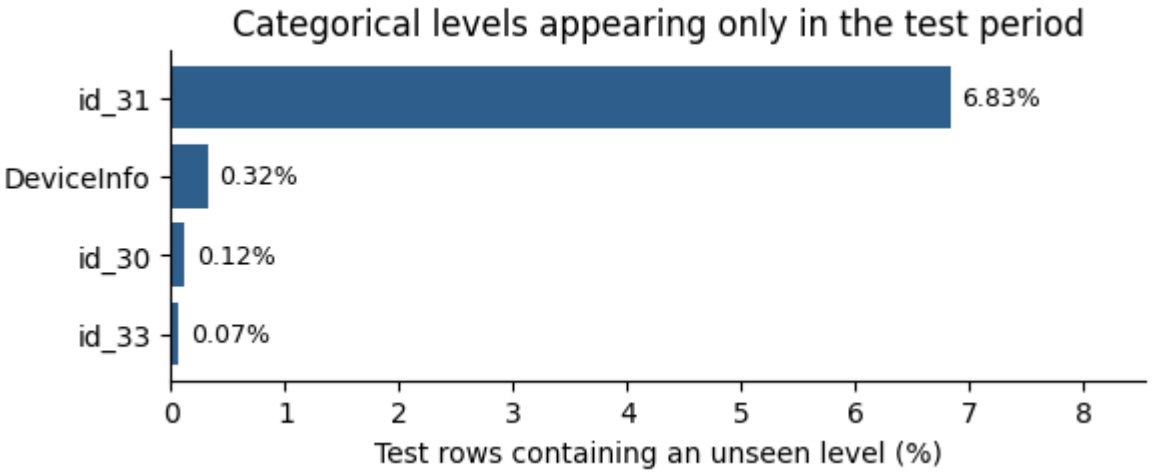


Figure 12. *Categorical columns containing levels present only in the test period, as a percentage of test rows.*

Only four categorical variables contain levels that are not observed in the training data. For three of them, these unseen levels account for less than 0.35% of the test observations. The exception is id_31, where the proportion reaches 6.83%. Since id_31 contains browser information, including version numbers, new values appearing in the later period are expected as browsers are updated. During preprocessing, unseen levels are encoded as −2 to distinguish them from missing values.

> **Implication** This also shows an effect of using a chronological split, since some categories appear only in the later test period and might be present in both sets under random splitting. Although the proportion is small for most variables, the result highlights the need for an encoding method that can handle categories not observed during training.

04 The Preprocessing Pipeline

The preprocessing pipeline was designed to prevent target and temporal information leakage while producing consistent, model ready datasets for the subsequent feature selection methods and the classification models. All preprocessing steps were fitted using only the training dataset. The resulting parameters, including numerical imputation values and categorical encoding rules, were then consistently applied to the test dataset to ensure a fair and consistent evaluation.

Step 1 — Load and merge

train_transaction.csv + train_identity.csv , Left join via TransactionID to retain all core transactions while appending sparse identity logs. And Drop TransactionID.

```
import numpy as np, pandas as pd

# 1. Load and merge
tx  = pd.read_csv('data/train_transaction.csv')
idn = pd.read_csv('data/train_identity.csv')
df  = tx.merge(idn, how='left', on='TransactionID')
del tx, idn
df = df.drop(columns=['TransactionID'])
```

Step 2 — Derive the transaction hour

```
# 2. TransactionDT is a seconds offset from a fixed reference point.

df["TransactionHour"] = ((df["TransactionDT"] // 3600) % 24)
```

Step 3 — Identify feature types, Numerical and Categorical columns

```
# 3. Identify feature types (excluding identifier and target)

TARGET = "isFraud"
num_cols = [c for c in df.select_dtypes(include=["number"]).columns
            if c not in (TARGET, "TransactionDT")]
cat_cols = list(df.select_dtypes(include=["object", "category"]).columns)
```

Step 4 — Chronological 80/20 split

Sort rows by ascending TransactionDT. Divide datasets sequentially to simulate historical training and out-of-sample future testing

```
df = df.sort_values("TransactionDT").reset_index(drop=True)
cut = int(len(df) * 0.80)
train = df.iloc[:cut].copy()   # earlier   (80%)
test  = df.iloc[cut:].copy()   # later (20%)
```

Step 5 — Numeric imputation

Compute medians on training data; apply to train and test.

```
train_medians = train[num_cols].median()
train[num_cols] = train[num_cols].fillna(train_medians)
test[num_cols]  = test[num_cols].fillna(train_medians)
```

Step 6 — Encode categorical

Map each categorical column to integers using training data only. Missing values are coded −1 and levels appearing only in the test period −2.

```
encoders = {}
for col in cat_cols:
    tr_na, te_na = train[col].isna(), test[col].isna()
    tr_s,  te_s  = train[col].astype(str), test[col].astype(str)

    levels  = sorted(tr_s[~tr_na].unique())
    mapping = {lvl: i for i, lvl in enumerate(levels)}

    train[col] = tr_s.map(mapping).where(~tr_na, -1).astype("int32")

    te_code = te_s.map(mapping).where(~te_na, -1)   # missing -> -1
    test[col] = te_code.fillna(-2).astype("int32")  # unseen  -> -2
    encoders[col] = mapping
```

Step 7 — Separate features and target

```
X_train = train.drop(columns=[TARGET]).copy()
y_train = train[TARGET].copy()
X_test  = test.drop(columns=[TARGET]).copy()
y_test  = test[TARGET].copy()
```

Chronological Modeling Partitions Summary

Partition	Rows	Fraud rate	Purpose
Training (earlier 80%)	472,432	3.51%	Fit preprocessing, feature selection, classifiers, and internal validation.
Test (later 20%)	118,108	3.44%	One-time final evaluation on future transactions.

❖ Class Imbalance Treatment Strategy

While oversampling techniques like SMOTE are commonly used to address class imbalance, they were deliberately excluded from this pipeline. SMOTE generates synthetic data points through interpolation, which can introduce artificial noise, distort true transactional distributions, and lead to over-optimistic performance evaluations in highly non-stationary fraud domains.

Instead, I will use in my study cost-sensitive learning by embedding class weights directly into each downstream machine learning classifier. This approach penalizes misclassifications of the minority fraud class during training without modifying the underlying empirical data distribution. The specific algorithmic implementations of these class weights are detailed in the accompanying feature

selection and classification scripts, while the final evaluation metrics are compiled and presented in the next file.

05 Conclusion

This report presented the exploratory data analysis (EDA) and preprocessing methodology developed for the high dimensional IEEE-CIS Fraud Detection dataset. The analysis identified the dataset's principal challenges, including severe class imbalance, extensive missing values, temporal dependencies, and a high-dimensional feature space.

The preprocessed data provide the foundation for the next stage of this research, which will investigate and compare multiple feature selection techniques. The selected feature subsets will subsequently be evaluated using several machine learning classifiers to identify the most effective approach for accurate and robust e-commerce fraud detection.

06 References and Data Sources

- IEEE-CIS Fraud Detection dataset — Kaggle competition; data provided by Vesta Corporation.
- Kaggle EDA reference notebook — “IEEE-CIS Fraud Detection: EDA / LightGBM” (external descriptive reference).
- Mohan, “IEEE-CIS Fraud Detection — Top 5% Solution,” Towards Data Science (Medium), 2021.
- Author’s working notebook — IEEE_CIS_Preprocessing_EDA.ipynb (reviewed 19 July 2026).